page 597/ Exercise 5

$\boxed{PN\ 597\ /\ EN\ 5}$

(a) $f(x) \geq 0$; $f(x) = \frac{c}{1+x^2}$, so $f(x) \geq 0$ when $c \geq 0$

$$\int_{-\infty}^{\infty} f(x)\,dx = 1$$

$$\int_{-\infty}^{\infty} \frac{c}{1+x^2}\,dx = 1 \Rightarrow c\int_0^{\infty} \frac{1}{1+x^2}\,dx = c \lim_{t\to\infty}\left[\tan^{-1} x\right]_0^t =$$

$$= c \lim_{t\to\infty} \tan^{-1} t = c\,\frac{\pi}{2};$$

as $\int_{-\infty}^{0} f(x)\,dx = \int_0^{\infty} f(x)\,dx \Rightarrow \int_{-\infty}^{\infty} f(x)\,dx = 2\int_0^{\infty} f(x)\,dx$

$\int_{-\infty}^{\infty} \frac{c}{1+x^2}\,dx = 2\int_0^{\infty} \frac{c}{1+x^2}\,dx = \cancel{2}c\,\frac{\pi}{\cancel{2}} = \pi c$ and it must be equal to $1 \Rightarrow \pi c = 1 \Rightarrow c = \frac{1}{\pi}$

(b)

$$P(-1 < X < 1) = \int_{-\infty}^{1} \frac{1/\pi}{1+x^2} = \frac{2}{\pi}\int_0^1 \frac{1}{1+x^2}\,dx = \frac{2}{\pi}\left[\tan^{-1} x\right]_0^1 =$$

$$= \frac{2}{\pi}\left(\frac{\pi}{4} - 0\right) = \frac{\cancel{2}}{\cancel{\pi}} \cdot \frac{\cancel{\pi}}{\cancel{4}} = \frac{1}{2}$$

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$\boxed{\text{Pr } 598 / \text{Exr } 11}$

$\mu = 2,5$ (~~minute~~)

(a) $\int_4^\infty \frac{2}{5} \cdot e^{-\frac{t2}{5}}\, dt = \lim_{x\to\infty} \frac{2}{5}\int_4^x e^{-\frac{2t}{5}}\, dt = \frac{2}{5}\left[-\frac{5}{2e^{\frac{2t}{5}}}\right]_4^x =$

$= \lim_{x\to\infty}\left(-\frac{1}{e^x}\right) - \left(-\frac{1}{e^{8/5}}\right) = e^{-\frac{8}{5}} \approx 0,202$ (20,2%)

(b) $\int_0^2 \frac{2}{5} \cdot e^{-\frac{t2}{5}}\, dt = \left[-\frac{1}{e^{2x/5}}\right]_0^2 = \left(-\frac{1}{e^{4/5}}\right) - (-1) = 1 - \frac{1}{e^{4/5}} \approx 0,551$

(55,1%)

(c) $\int_x^\infty \frac{2}{5} e^{-\frac{2t}{5}}\, dt = 0,02$

$\lim_{y\to\infty}\int_x^y \frac{2}{5} e^{-\frac{2t}{5}}\, dt = 0,02$

$\lim_{y\to\infty}\left[-e^{-\frac{2t}{5}}\right]_x^y = \left(-\frac{1}{e^y}\right) - \left(-e^{-2x/5}\right) = \lim_{y\to\infty}\left(-\frac{1}{e^y}\right) + e^{-2x/5} = 0,02$

$e^{-2x/5} = 0,02$

$-2x/5 = \ln\frac{1}{50}$

$x = -\frac{5}{2}\ln\frac{1}{50} \approx 9,78$ (minute)

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P№598/E№12

$\mu = 69$

$\sigma = 2{,}8$

(a) $\int_{65}^{73} \frac{1}{2{,}8\sqrt{2\pi}} \cdot e^{-(x-69)^2/(2\cdot 2{,}8^2)}\,dx \approx 0{,}847$ (84,7%)

(b) $1 - \int_{0}^{72} \frac{1}{2{,}8\cdot\sqrt{2\pi}} \cdot e^{-(x-69)^2/(2\cdot 2{,}8^2)}\,dx \approx 0{,}142$ (14,2%)

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P№598/E№15

speed limit = 100 km/h

$\mu = 112$ km/h

$\sigma = 8$ km/h

(a) $\int_{0}^{100} \frac{1}{8\sqrt{2\pi}} \cdot e^{-(x-12)^2/(2\cdot 8^2)}\,dx \approx 0{,}0668$ (6,68%)

(b) $1 - \int_{0}^{125} \frac{1}{8\sqrt{2\pi}} \cdot e^{-(x-112)^2/(2\cdot 8^2)}\,dx \approx 0{,}052$ (5,2%)